

# Mengran Li

PhD Candidate in Statistics

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## Summary

I develop causal inference methodology designed for rare and extreme events. My work contributes to causal discovery, estimation, and attribution in settings involving extreme events and extremal dependence. By synthesizing causal frameworks with extreme value theory, I aim to enable reliable inference in high-impact extreme regimes with a focus on climate risk assessment.

## Education

### University of Glasgow

Doctor of Philosophy in Statistics

Glasgow, UK

2022 - 2026

### University of Glasgow

Master of Science in Statistics with distinction

Glasgow, UK

2021 - 2022

### Southwestern University of Finance and Economics

Bachelor of Science in Statistics (Major)

Bachelor of Science in Finance (Minor)

Chengdu, China

2015 - 2019

## Publications

\* denotes corresponding author

1. **Li, M.\*** & Castro-Camilo, D. High-Dimensional Causal Discovery in Multivariate Extremes. *Manuscript in preparation*.
2. **Li, M.\*** & Castro-Camilo, D. (2026). Tail-Calibrated Estimation of Extreme Quantile Treatment Effects. *In preparation*.
3. **Li, M.\*** & Castro-Camilo, D. (2025). On the importance of tail assumptions in climate extreme event attribution. *arXiv:2507.14019*. Under review at *Climatic Change*.
4. **Li, M.\***, Cuba, D., Hu, C., & Castro-Camilo, D. (2024). A wee exploration of techniques for risk assessments of extreme events. *Extremes*. doi:10.1007/s10687-024-00500-5
5. **Li, M.\*** & Castro-Camilo, D. (2025). eFCM: An R package for the Exponential Factor Copula Model. CRAN R package. [cran.r-project.org/package=eFCM](https://cran.r-project.org/package=eFCM)

## Research Experience

### Causal Structure Learning for Multivariate Extremes

Developed structure learning methods for high-dimensional multivariate extremes and tail dependence, enabling recovery of directional causal structure encoded by directed acyclic graphs in extreme regimes.

Glasgow, UK

2025 - 2026

### Causal Inference for Extreme Quantile Treatment Effects

Developed a causal inference framework based on inverse estimating equations for estimating treatment effects in the far tails of outcome distributions, with formal identification and asymptotic guarantees.

Glasgow, UK

2023 - 2025

### Tail Dependence and Causal Attribution of Climate Extremes

Integrated multivariate extreme value modeling into a counterfactual causal inference framework to assess the role of tail dependence in extreme event attribution, showing that dependence assumptions critically shape causal conclusions.

Glasgow, UK

2022 - 2025

Teaching Experience

|                                  |                                                                                                                                                                                    |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>University of Glasgow</b>     | Glasgow, UK                                                                                                                                                                        |
| Tutor and Demonstrator           | 2022 – 2026                                                                                                                                                                        |
| <b>Statistics &amp; Modeling</b> | Bayesian Statistics, Time Series, Flexible Regression, Biostatistics, Advanced Predictive Models                                                                                   |
| <b>Statistical Computing</b>     | Introduction to R & Python, Data Analysis                                                                                                                                          |
| <b>Responsibilities</b>          | Led tutorials and lab sessions for undergraduate and MSc students; provided technical support for programming assignments; assessed coursework and provided constructive feedback. |

Conference Presentations

- First workshop of the Glasgow-Edinburgh Extremes Network**, 11 December 2025, University of Glasgow, Glasgow (UK). *Talk presentation.*
- ISI World Statistics Congress 2025**, 5–9 October 2025, International Statistical Institute, The Hague (Netherlands). *Talk presentation.*
- RSS 2025 International Conference**, 1–4 September 2025, Royal Statistical Society, Edinburgh (UK). *Poster presentation.*
- The 6th International Conference on Advances in Extreme Value Analysis and Application to Natural Hazards (EVAN 2024)**, 16–19 July 2024, Istituto Veneto di Scienze, Venice (Italy). *Talk presentation.*
- STOR-i Extremes Workshop**, 20–22 September 2023, Lancaster University, Lancaster (UK). *Talk presentation.*
- Research Students’ Conference in Probability and Statistics**, 11–14 September 2023, University of Sheffield, Sheffield (UK). *Talk presentation; Travel grant recipient.*

Awards and Recognitions

- Google Cloud Research Grant.** December 2025.
- 3rd Place, ISAC Data Analysis Competition, Travel grant awarded.** ISI World Statistics Congress, The Hague (Netherlands), July 2025.
- Fully funded PhD scholarship.** 2022–2026.
- Academic Scholarships.** Southwestern University of Finance and Economics. 2015–2019.

Academic Service

|                                                                                                                                                                                                                               |                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>Journal Reviewer</b>                                                                                                                                                                                                       | 2025 - Present |
| Data-Centric Engineering (Cambridge University Press)                                                                                                                                                                         |                |
| <b>Seminar &amp; Reading Group Organisation</b>                                                                                                                                                                               |                |
| Co-organized the monthly seminar series of the <b>Glasgow–Edinburgh Extremes Network (GLE<sup>2</sup>N)</b> , coordinating invited speakers and supporting seminar delivery for sessions hosted at the University of Glasgow. | 2025 - Present |

Skills

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|--------------------------------------------------|----------------------------------------------------------------------------|
| <b>Programming &amp; Computing</b>               | R (advanced), Python, C++, Linux, Git, HPC, Cloud Computing (Google Cloud) |
| <b>Scientific Software &amp; Reproducibility</b> | R package development (CRAN), R Markdown, LaTeX                            |
| <b>Web &amp; Documentation</b>                   | HTML/CSS, Hugo                                                             |